

SYBTech AIML - Inferential Statistics

Hypothesis Testing & Non-parametric Methods

SECTION A: 6 MARKS QUESTIONS

- Q1.** Define Most Powerful (MP) Test. State and prove Neyman-Pearson Lemma.
- Q2.** Explain Uniformly Most Powerful (UMP) Test. Discuss its relationship with MP Test and provide an example.
- Q3.** Derive the Likelihood Ratio Test for testing $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$ for a normal population with known variance σ^2 .
- Q4.** State the properties of Likelihood Ratio Test (LRT). Explain why LRT is preferred over MP Test in practice.
- Q5.** Derive the test statistic for testing $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 \neq \mu_2$ for two independent normal populations with equal variances. Give the critical region.
- Q6.** Explain Analysis of Variance (ANOVA). Derive the F-statistic for testing equality of means of k normal populations.
- Q7.** Derive the test statistic for testing $H_0: \sigma^2 = \sigma_0^2$ vs $H_1: \sigma^2 \neq \sigma_0^2$ for a normal population. Discuss the assumptions required.
- Q8.** Develop a test for equality of variances of two normal populations. Derive the F-statistic and discuss its distribution.
- Q9.** Compare parametric and non-parametric methods. Discuss advantages and disadvantages of non-parametric tests.
- Q10.** Explain the Mann-Whitney U Test. When is this test preferred over t-test? Derive the test statistic.
- Q11.** Describe the Wilcoxon Signed-Rank Test. When is this test used and what are its advantages over paired t-test?
- Q12.** Discuss the Kruskal-Wallis Test. How does it differ from ANOVA and when should it be used?

SECTION B: 5 MARKS QUESTIONS

- Q1.** Explain the concept of critical region and power function in hypothesis testing. How are they related to Type I and Type II errors?
- Q2.** Derive the LRT for $H_0: p = p_0$ vs $H_1: p \neq p_0$ where p is the probability of success in Binomial distribution.
- Q3.** A sample of 25 items from a normal population gives mean 16 and standard deviation 2.5. Test $H_0: \mu = 15$ vs $H_1: \mu \neq 15$ at 5% level.
- Q4.** Two samples of sizes 20 and 18 from normal populations give means 25 and 22 with variances 16 and 12 respectively. Test for equality of means at 5% level.
- Q5.** Explain the F-test for testing equality of variances. A sample of 10 observations from population 1 gives variance 24, and 12 observations from population 2 gives variance 18. Test at 10% level.
- Q6.** Discuss the advantages and limitations of non-parametric tests compared to parametric tests.
- Q7.** Explain the Sign Test and its application. When is it used instead of parametric tests?
- Q8.** Describe the Spearman's Rank Correlation Coefficient. How is it calculated and when is it preferred over Pearson's correlation?
- Q9.** Explain the Chi-square Goodness-of-Fit Test. How is it used to test whether data follows a specific distribution?
- Q10.** Define confidence intervals and their relationship with hypothesis testing. How are they constructed for means and variances?
- Q11.** Explain sequential analysis in hypothesis testing. What are its advantages in quality control and clinical trials?
- Q12.** Discuss the concept of p-value. How is it used in hypothesis testing and what are common misconceptions about it?

SECTION C: 4 MARKS QUESTIONS

- Q1.** Define Type I and Type II errors with examples. Explain the relationship between α and β .
- Q2.** State Neyman-Pearson Lemma and explain its significance.
- Q3.** Explain the concept of power of a test and how it increases with sample size.
- Q4.** Derive the test statistic for testing $H_0: \mu = \mu_0$ vs $H_1: \mu > \mu_0$ for normal population with unknown variance.
- Q5.** Explain the t-test for single mean. When is it used instead of z-test?
- Q6.** Describe the paired t-test. Give an example where paired t-test is applicable.
- Q7.** Explain ANOVA one-way classification. What is the null and alternative hypothesis?
- Q8.** Define the F-distribution and explain its role in variance testing.
- Q9.** Explain the chi-square test for independence. How is the test statistic calculated?
- Q10.** State the advantages of using median test over mean test in non-normal distributions.
- Q11.** Explain the Mood's Median Test and when it is preferred.
- Q12.** Describe the run test and its application in testing randomness.
- Q13.** Explain the Kolmogorov-Smirnov Test for goodness of fit.
- Q14.** Define orthogonal contrasts and their role in ANOVA.
- Q15.** Explain the concept of multiple testing and the need for correction.
- Q16.** Discuss Bonferroni correction and its application in multiple hypothesis testing.
- Q17.** Explain the assumptions required for validity of t-test.
- Q18.** When should we use non-parametric tests instead of parametric tests?
- Q19.** Describe the effect size and its importance in hypothesis testing.
- Q20.** Explain the relationship between confidence interval and hypothesis testing.

SECTION D: NUMERICAL PROBLEMS (6 MARKS)

Q1. A sample of 30 students has mean IQ of 105 and standard deviation 12. Test whether the population mean IQ is 100 at 5% level of significance.

Q2. Two groups of students were given different teaching methods. Group 1 ($n=25$): mean=78, $s=8$. Group 2 ($n=20$): mean=72, $s=10$. Test for equality of means at 5% level.

Q3. The tensile strength of steel samples from two suppliers gave: Supplier A ($n=15$): mean=500, variance=25. Supplier B ($n=12$): mean=495, variance=36. Test for equality of means at 1% level.

Q4. Data on heights (in cm) of 10 students: 165, 168, 170, 172, 175, 178, 180, 182, 185, 190. Test if mean height is 175 cm at 5% level.

Q5. Scores of 8 students in two subjects: Math: 78, 82, 85, 88, 90, 92, 95, 98. Science: 80, 84, 88, 85, 92, 90, 93, 96. Test if means are equal using paired t-test at 5% level.

NUMERICAL PROBLEMS (5 MARKS)

Q1. Three machines are used to produce components. Test if they produce components with equal mean weight: Machine A ($n=5$, mean=105, variance=4), Machine B ($n=6$, mean=107, variance=5), Machine C ($n=5$, mean=103, variance=6). Use $\alpha=0.05$.

Q2. A random sample of 25 defective items from a production process has mean lifetime of 800 hours with s.d. 50 hours. Test if population mean lifetime is 820 hours at 5% level.

Q3. Two quality inspectors tested 100 items each. Inspector A found 8 defective, Inspector B found 5 defective. Test if both inspectors have equal detection rates at 5% level.

Q4. Blood pressure readings of 12 patients before and after medication: Before: 150, 155, 160, 165, 158, 162, 168, 170, 155, 160, 165, 170. After: 140, 145, 150, 155, 148, 152, 158, 160, 145, 150, 155, 160. Test effectiveness of medication at 5% level.

Q5. A company claims their battery has mean life of 100 hours. Test this claim using a random sample of 16 batteries with mean=98 hours and s.d.=4 hours at 5% level.

NUMERICAL PROBLEMS (4 MARKS)

Q1. Test if the proportion of defective items in a batch is 0.05 using a sample of 200 items with 15 defective at 5% level.

Q2. Variance of 20 measurements from a process is 16. Test if population variance is 20 at 5% level.

Q3. Sample of 30 items gives mean=50, s.d.=8. Construct 95% confidence interval for population mean.

Q4. Two samples: $n_1=16$, $s_1^2=4$ and $n_2=12$, $s_2^2=5$. Test equality of variances at 5% level.

Q5. In a sample of 100 voters, 45 support a policy. Test if population proportion is 0.5 at 5% level.

Q6. Weekly sales data for 8 weeks: 100, 110, 105, 115, 120, 125, 118, 130. Test if mean sales is 110 at 5% level.

Q7. In a binomial experiment with $n=50$, $p=0.4$. Test $H_0: p=0.5$ vs $H_1: p \neq 0.5$ at 5% level.

Q8. Construct 90% confidence interval for difference of two population means using: $n_1=25$, $\bar{x}_1=60$, $s_1^2=8$, $n_2=20$, $\bar{x}_2=55$, $s_2^2=9$.